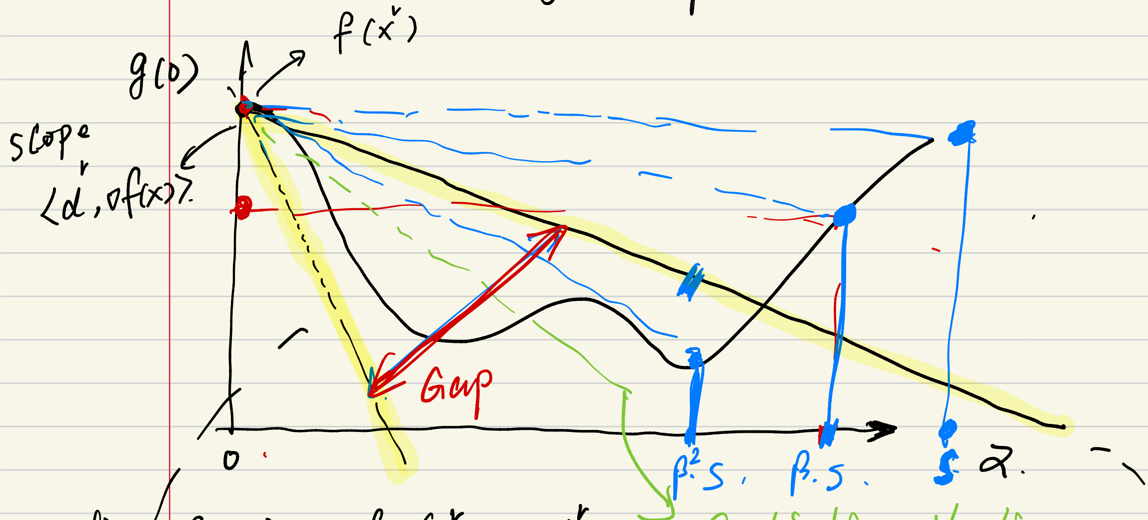
9/19/2024 lecture 2.

About the Armijo stepsize rule



Define $g(\alpha) := f(x^r + \alpha d^r)$

$g(0) = f(x^r)$

$g'(\alpha) = f'(x^r + \alpha d^r) = d^{r^T} \nabla f(x^r + \alpha d^r)$

$g'(0) = d^{r^T} \nabla f(x^r) = \langle d^r, \nabla f(x^r) \rangle < 0$

slope of any blue dashed line (for a given α')

$$\frac{g(\alpha') - g(0)}{\alpha' - 0} = \frac{g(\alpha') - g(0)}{\alpha'}$$

slop of black dashed line $\langle \nabla f(x^r), d^r \rangle$

slop of black solid line $\sigma \langle \nabla f(x^r), d^r \rangle$

(note, $\sigma < 1$)

So the Amijo rule process works the following.

① we know $\langle d, \nabla f(x) \rangle < 0$.

② Fix $\sigma \in (0, \frac{1}{4})$, then the solid black line is fixed with slop $\sigma \langle d, \nabla f(x) \rangle$

start setting $\alpha' = s$.

③ check if the blue line falls between the black dotted and black solid line by checking

$$\frac{g(\alpha') - g(0)}{\alpha' - 0} < \sigma \langle d, \nabla f(x) \rangle \quad (*)$$

If (*) satisfied, stop.

If not, then set $\alpha' = s \cdot \beta$, effectively rotate the blue line clockwise a bit.

Then go to ③ again

Note: since we know that there is a gap between the black line & the dotted black line, the above process must stop after finite steps